CS 312: FINAL PROJECT PRESENTATION

CLASSIFICATION & CLUSTERING WITH THE COVID-19 DATASET

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TABLE OF CONTENTS

Presentation Outline

ABOUT THE DATASET

SPECIAL SURVEY FOR EVALUATING SOCIO- ECONOMIC IMPACT OF COVID-19 ON WELLBEING OF PEOPLE

The dataset used in this project comes from the Special Survey for Evaluating the Socio-Economic Impact of COVID-19 on Wellbeing of People, conducted by the **Pakistan Bureau of Statistics (PBS)** under the **Ministry of Planning, Development & Special Initiatives.** It is one of the most comprehensive national household surveys capturing the multi-dimensional effects of COVID-19 across Pakistan.

ABOUT THE DATASET

Key Themes:

1. Demographics & Household Composition
2. Employment & Income Changes
3. Food Insecurity Conditions
4. Social Protection & Government Assistance
5. Housing & WASH Facilities
6. Coping Strategies During the Crisis
7. Health Services

PROBLEM DEFINITION

Predicting Socioeconomic Vulnerability During Calamities

COVID-19 affected households unevenly in Pakistan, with vulnerable families facing disproportionate income shocks, job losses, and food insecurity. The challenge is **to identify which households are most at risk during future crises** so that support programs can be targeted proactively.

1. **Food Insecurity:** Whether a household became food insecure during a crisis.
2. **Major Income Loss:** Whether a household experienced a significant drop in income during a crisis.

DATA UNDERSTANDING & EXPLORATION

Raw Data:

- ~30,000 individuals from ~5500 households sampled across all provinces.
- 329 features/attributes: demographics, employment, income changes, food insecurity, coping strategies, aid, health service access.
- Nationally representative, using stratified sampling across urban and rural regions.
- Mixed data types: numerical, categorical, binary Y/N indicators.

DATA UNDERSTANDING & EXPLORATION

Data Cleaning:

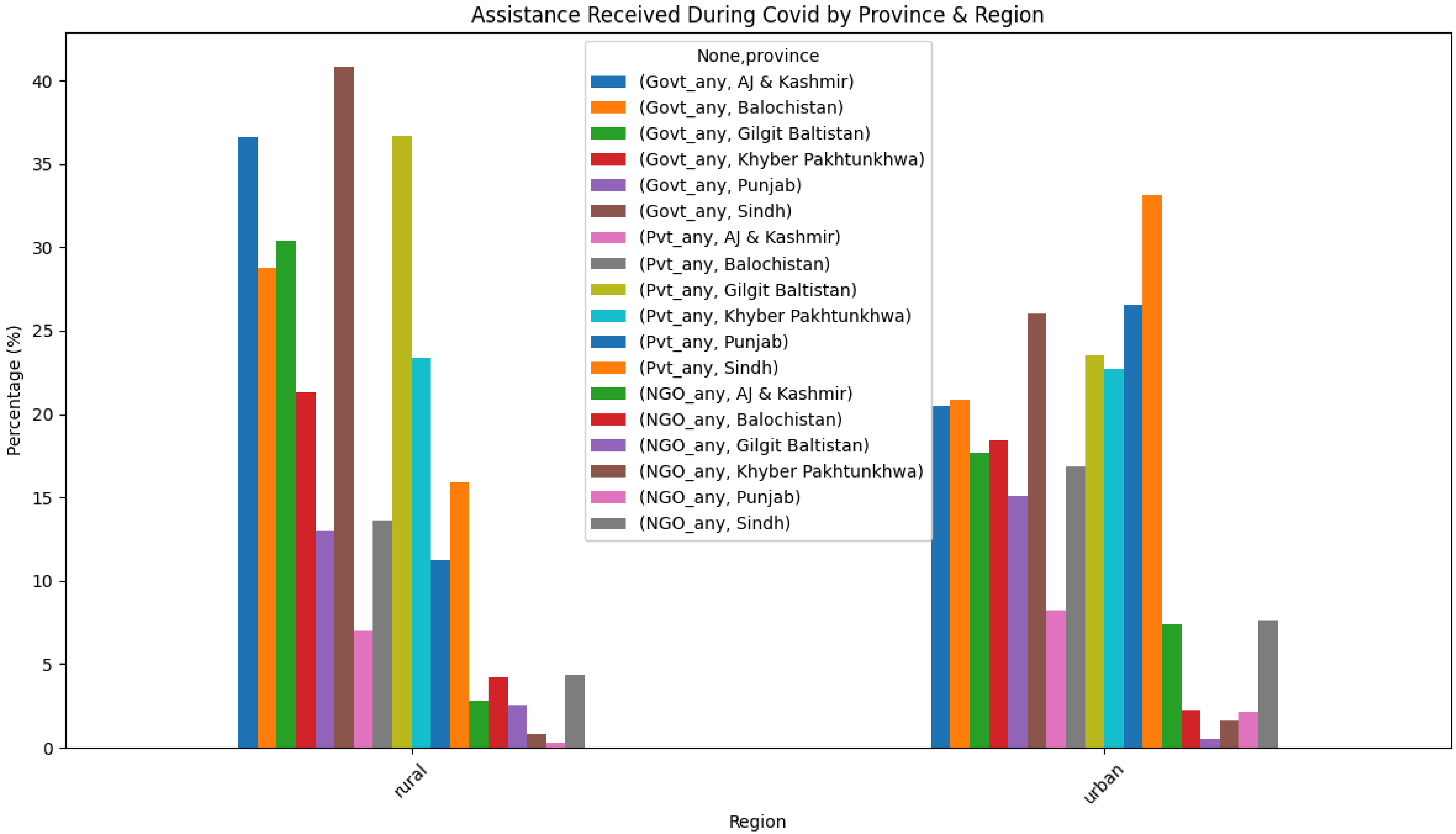
- Removed irrelevant administrative and survey-design columns
- Standardized "Don't Know", "Refused", blanks, and whitespaces
- Ensured numeric fields (income, rooms, household size, etc.) converted to proper datatypes
- Dropped columns with no analytical or predictive value for our goal

DATA UNDERSTANDING & EXPLORATION

Feature Engineering:

- Created derived socioeconomic indicators from each Section A-J
- Alot of pre-processing involved in this step

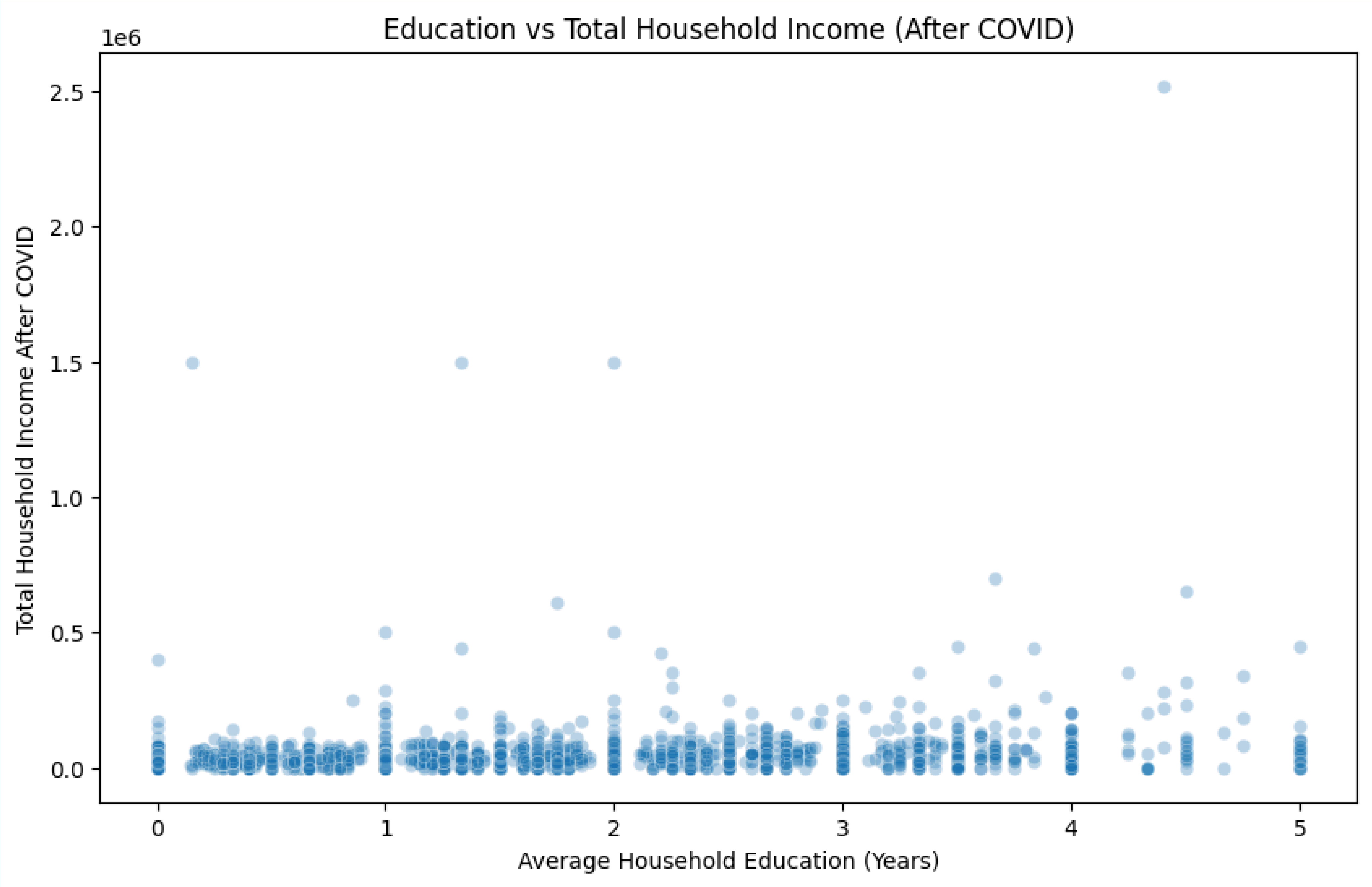
HOUSEHOLDS RECEIVING ASSISTANCE BY REGION



Overall trends:

- Rural Dominance
- Sindh and AJ&K (Rural) show the highest dependence on Government and NGO assistance.
- Sindh (Urban) shows the highest dependence on Private assistance.

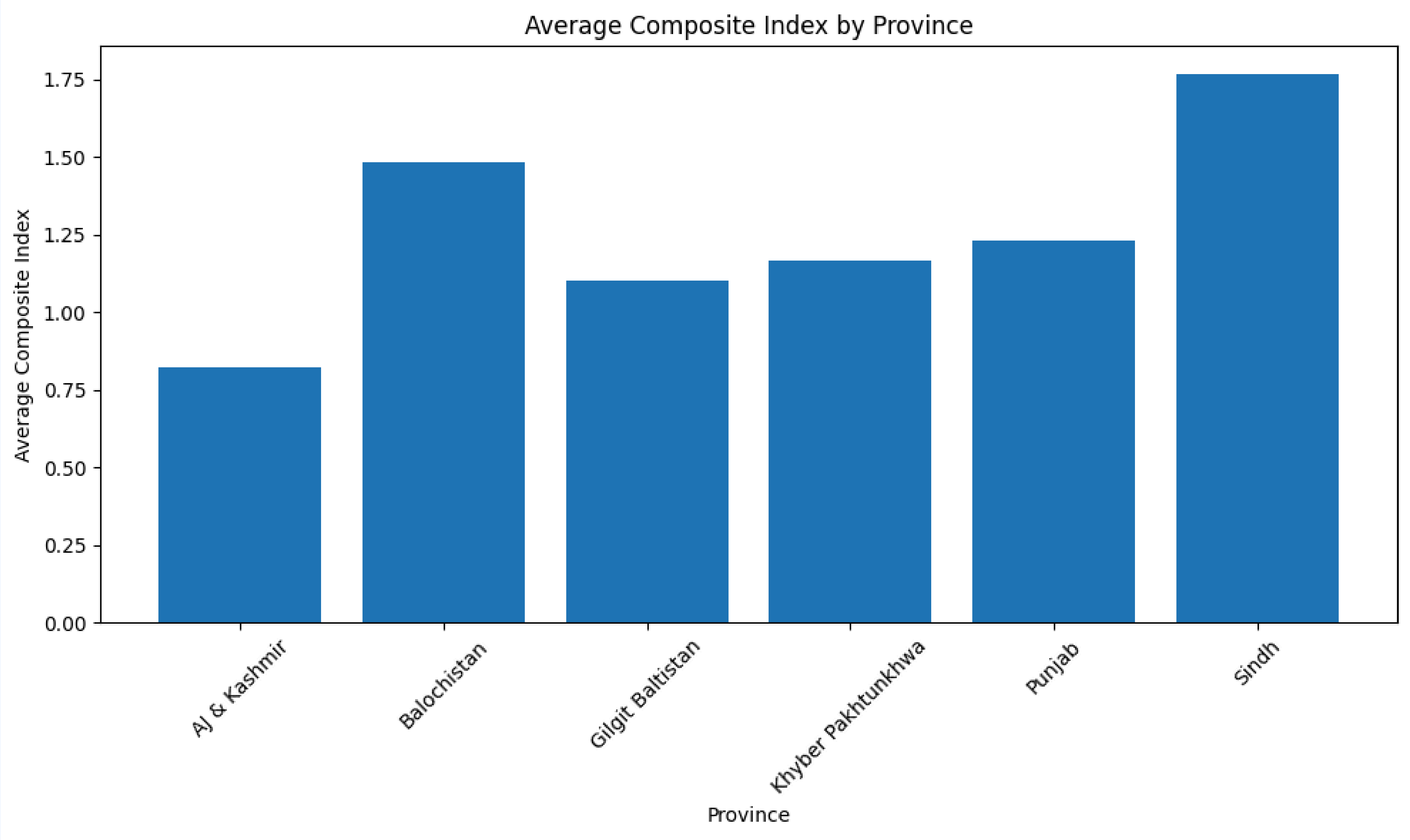
EDUCATION VS TOTAL HOUSEHOLD INCOME (AFTER COVID)



Low/high education does not guarantee low/high income.

However, the highest possible incomes (the income ceiling) are overwhelmingly restricted to households with higher average education levels

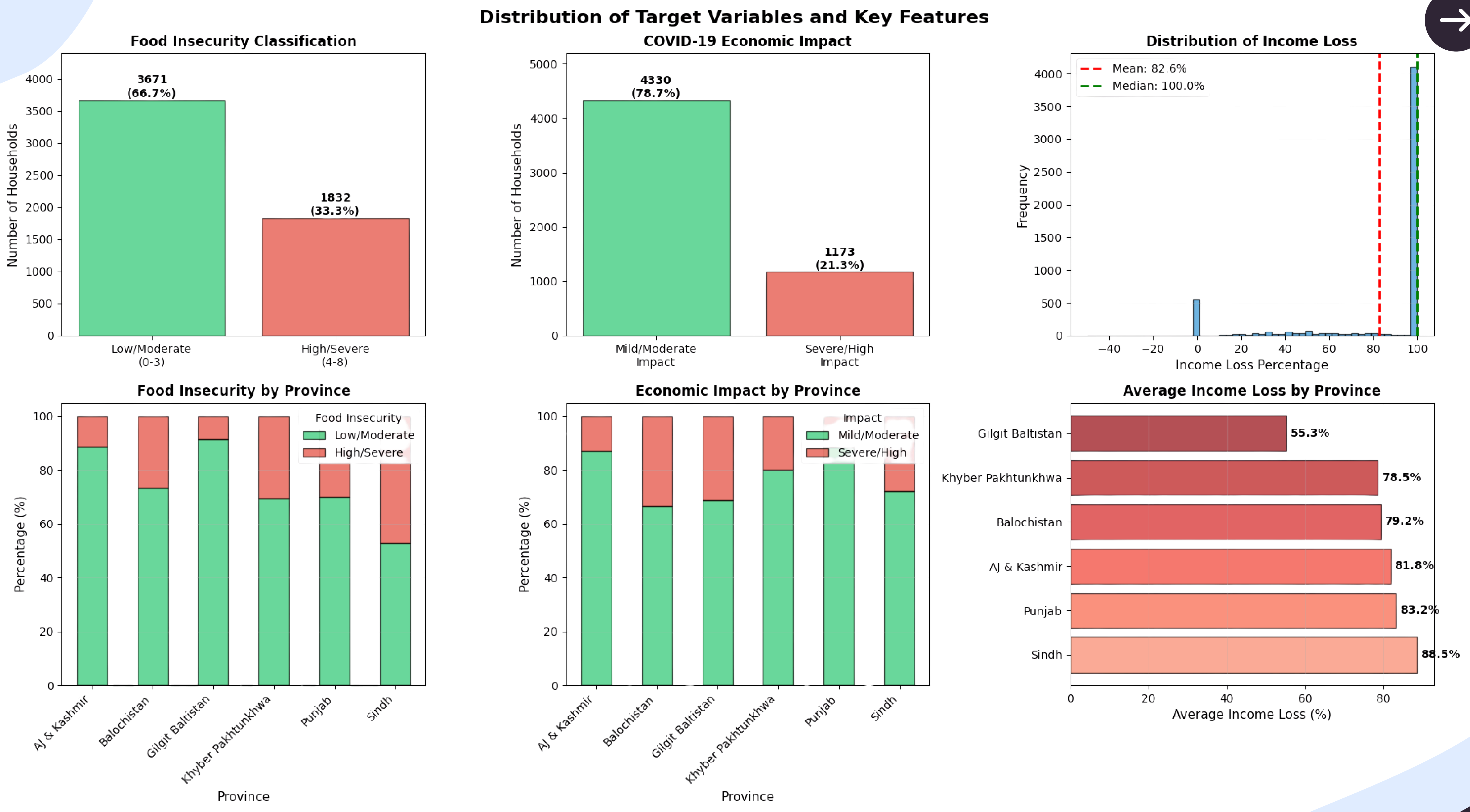
MOST BURDENED REGION



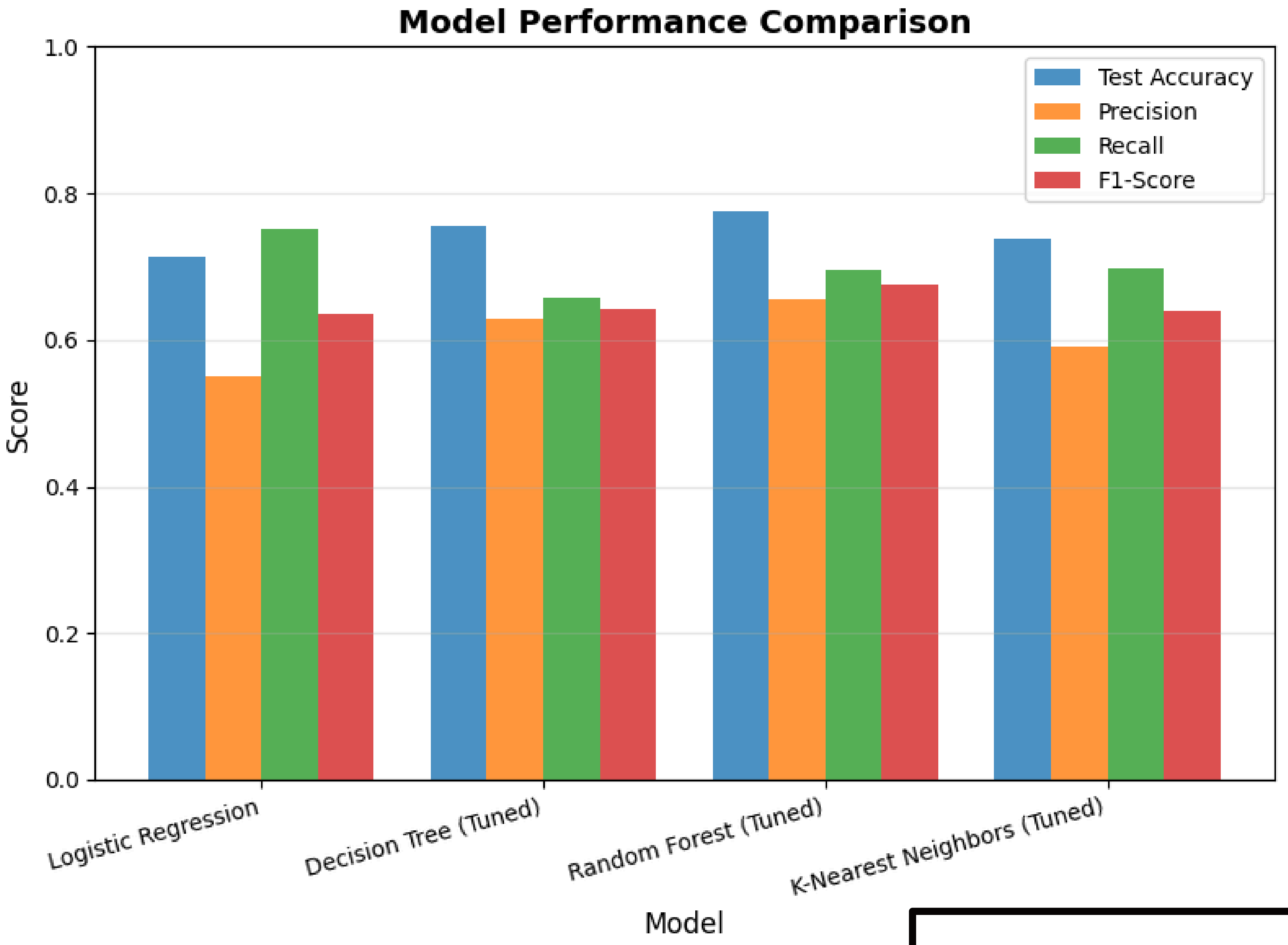
Sindh records the highest composite vulnerability score, indicating that households in Sindh faced the greatest combined burden of job losses, food insecurity, coping stress, and reduced access to health services during COVID-19

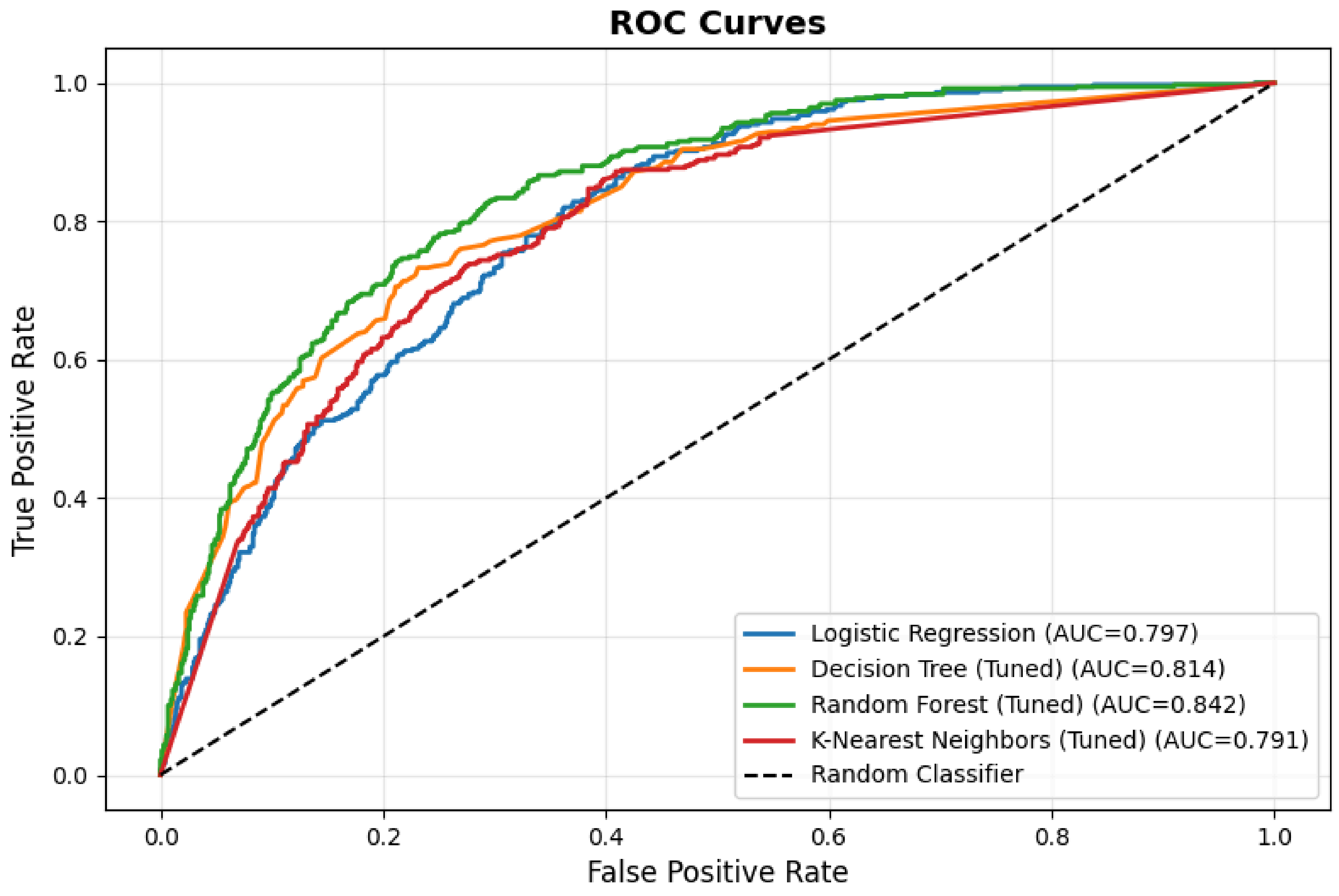
MODELING & PREDICTION

- K-Means Clustering (Unsupervised Learning)
- Classification Models
    - Target 1: Food Insecurity
    - Target 2: Economic Impact
- Regression Models
    - Target 3: Income Loss Prediction


Distribution of Target Variables and Key Features
Food Insecurity Classification
Number of Households
3671
(66.7%)
1832
(33.3%)
Low/Moderate
(0-3)
High/Severe
(4-8)
COVID-19 Economic Impact
4330
(78.7%)
1173
(21.3%)
Mild/Moderate
Impact
Severe/High
Impact
Distribution of Income Loss
Mean: 82.6%
Median: 100.0%
Frequency
Income Loss Percentage
Food Insecurity by Province
Food Insecurity
Low/Moderate
High/Severe
Percentage (%)
AJ & Kashmir
Balochistan
Gilgit Baltistan
Khyber Pakhtunkhwa
Punjab
Sindh
Province
Economic Impact by Province
Impact
Mild/Moderate
Severe/High
Average Income Loss by Province
Gilgit Baltistan
55.3%
Khyber Pakhtunkhwa
78.5%
Balochistan
79.2%
AJ & Kashmir
81.8%
Punjab
83.2%
Sindh
88.5%
Average Income Loss (%)

TARGET 1 - FOOD INSECURITY



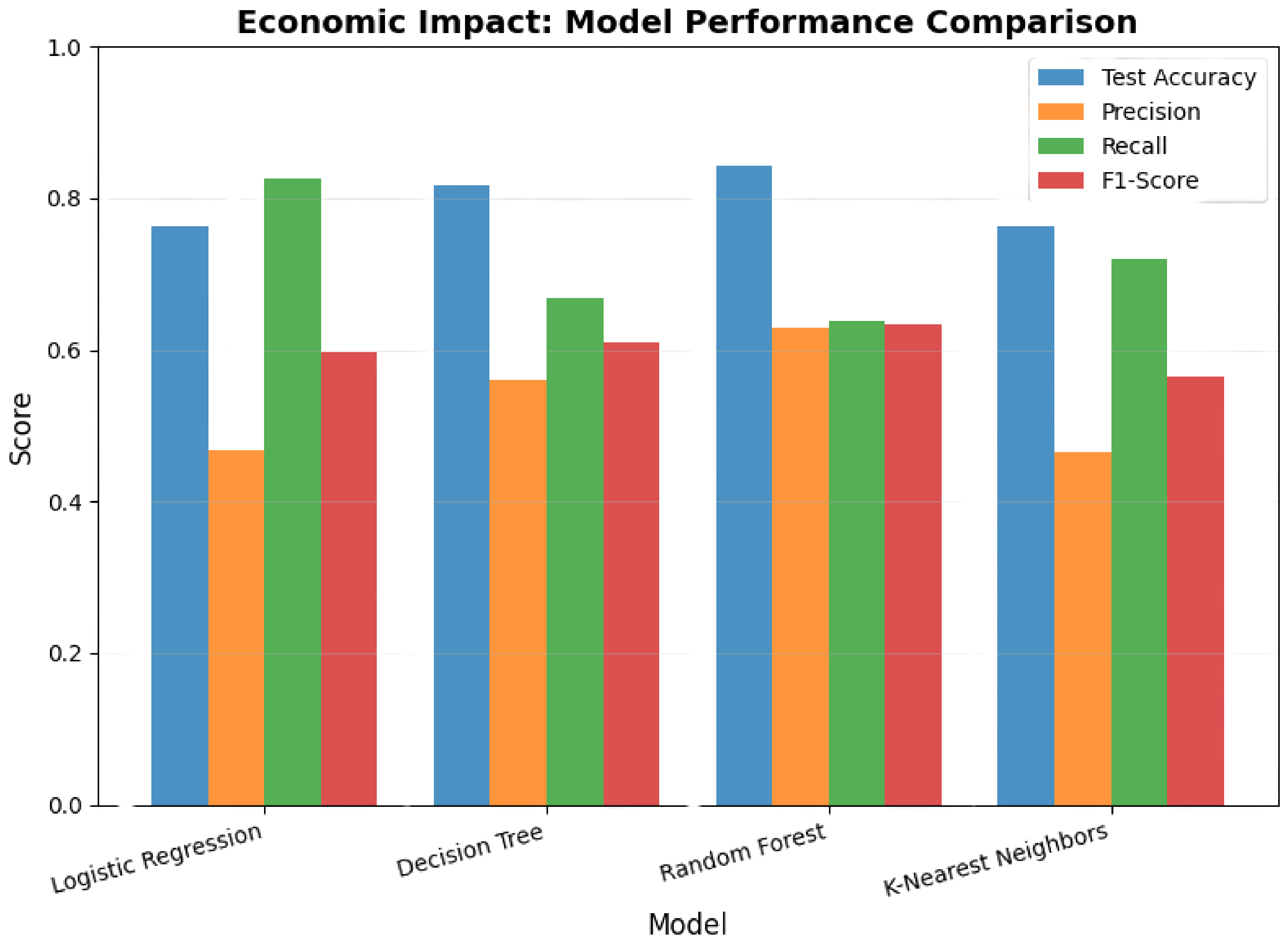


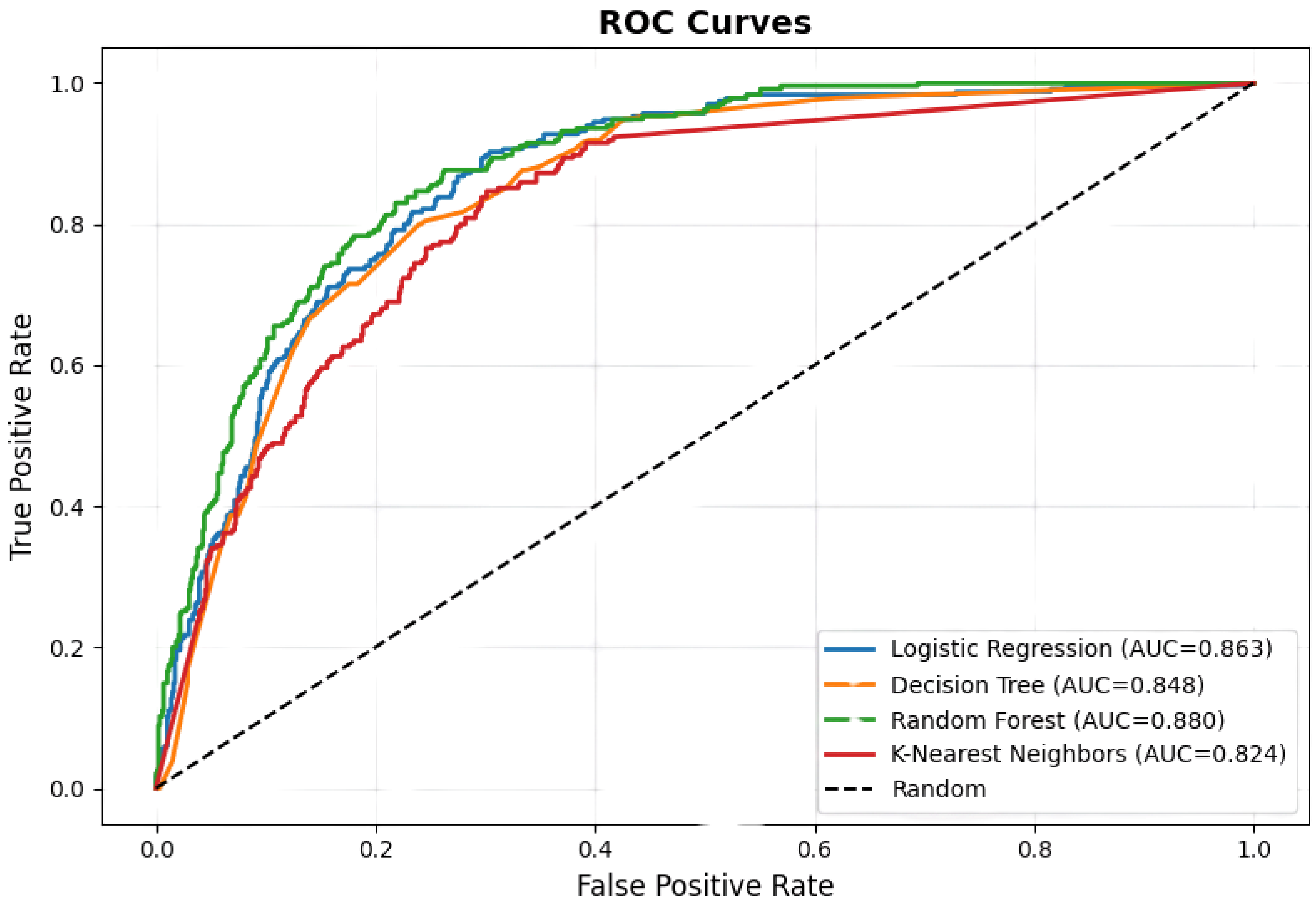
- Best Model
  - Random Forest
  - Test Accuracy: 77.66%
  - F1-Score: 0.6746

Most Important Predictors:

- Coping mechanisms used
- Income loss percentage
- Education level
- Government assistance received

TARGET 2 – ECONOMIC IMPACT





Best Model:

- Random Forest
- Test Accuracy: 84.29%
- F1-Score: 0.6342

TARGET 3 – INCOME LOSS PREDICTION

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REGRESSION MODEL COMPARISON - INCOME LOSS PREDICTION
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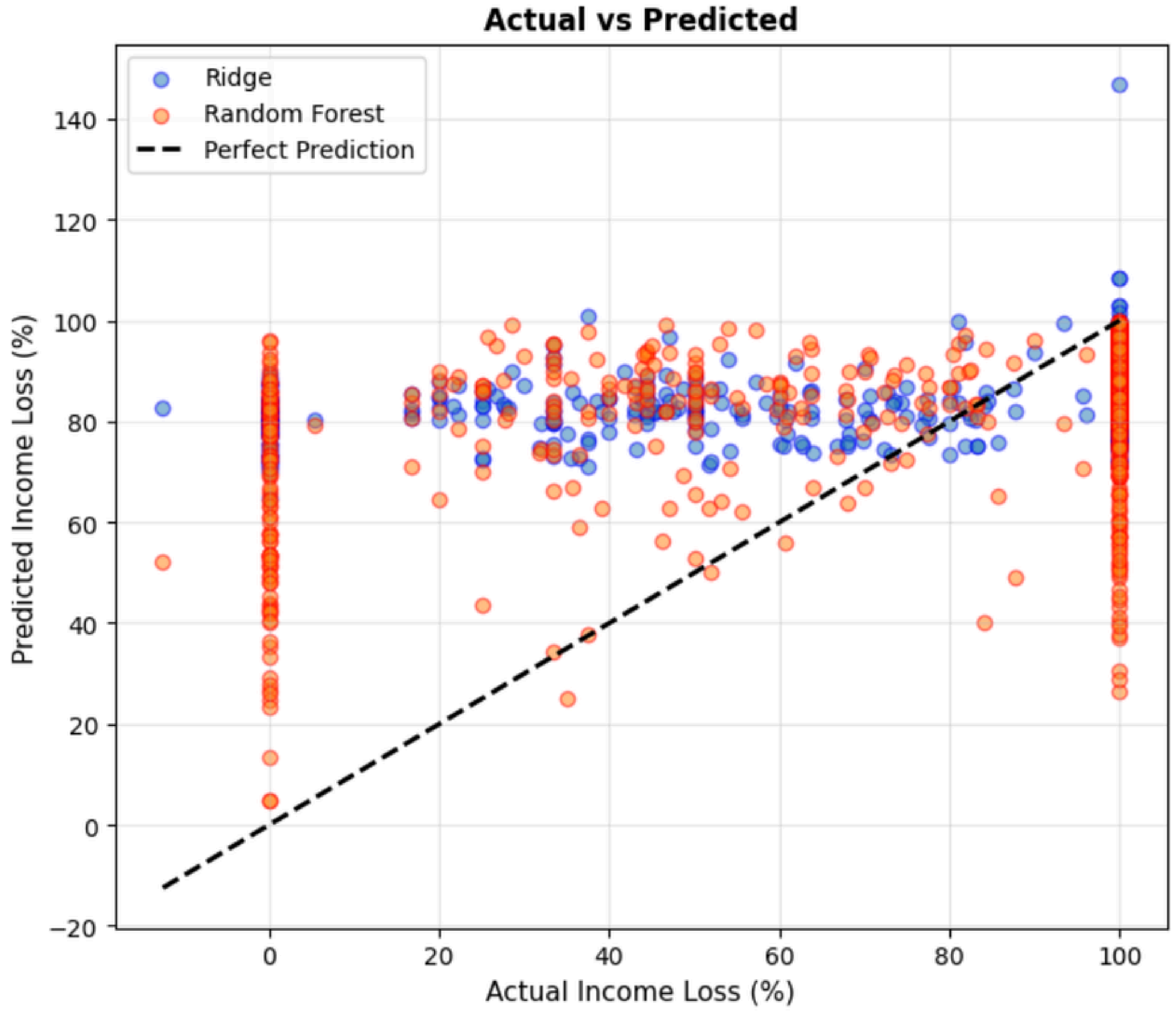
                    Model  Train R²  Test R²  MAE (%)  RMSE (%)
         Ridge Regression    0.0330   0.0356  26.1253   33.9798
  Random Forest Regressor    0.3731   0.1618  23.1235   31.6801
```



TARGET 3 - INCOME LOSS PREDICTION (HYBRID APPROACH)

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[STEP 1] Classification: Income Loss = 100% or Not
------------------------------------------------------
Training samples with 100% loss: 3316 (75.5%)
Training samples with <100% loss: 1074 (24.5%)

Classifier Accuracy: 0.723

Classification Report:
              precision    recall  f1-score   support

       <100%       0.51      0.24      0.33       308
        100%       0.75      0.91      0.83       790

    accuracy                           0.72      1098
   macro avg       0.63      0.58      0.58      1098
weighted avg       0.69      0.72      0.69      1098
```

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Comparison:
  Ridge:  R² = 0.0356, MAE = 26.13%
  Random Forest: R² = 0.1618, MAE = 23.12%
  Hybrid: R² = -0.2566, MAE = 19.94%
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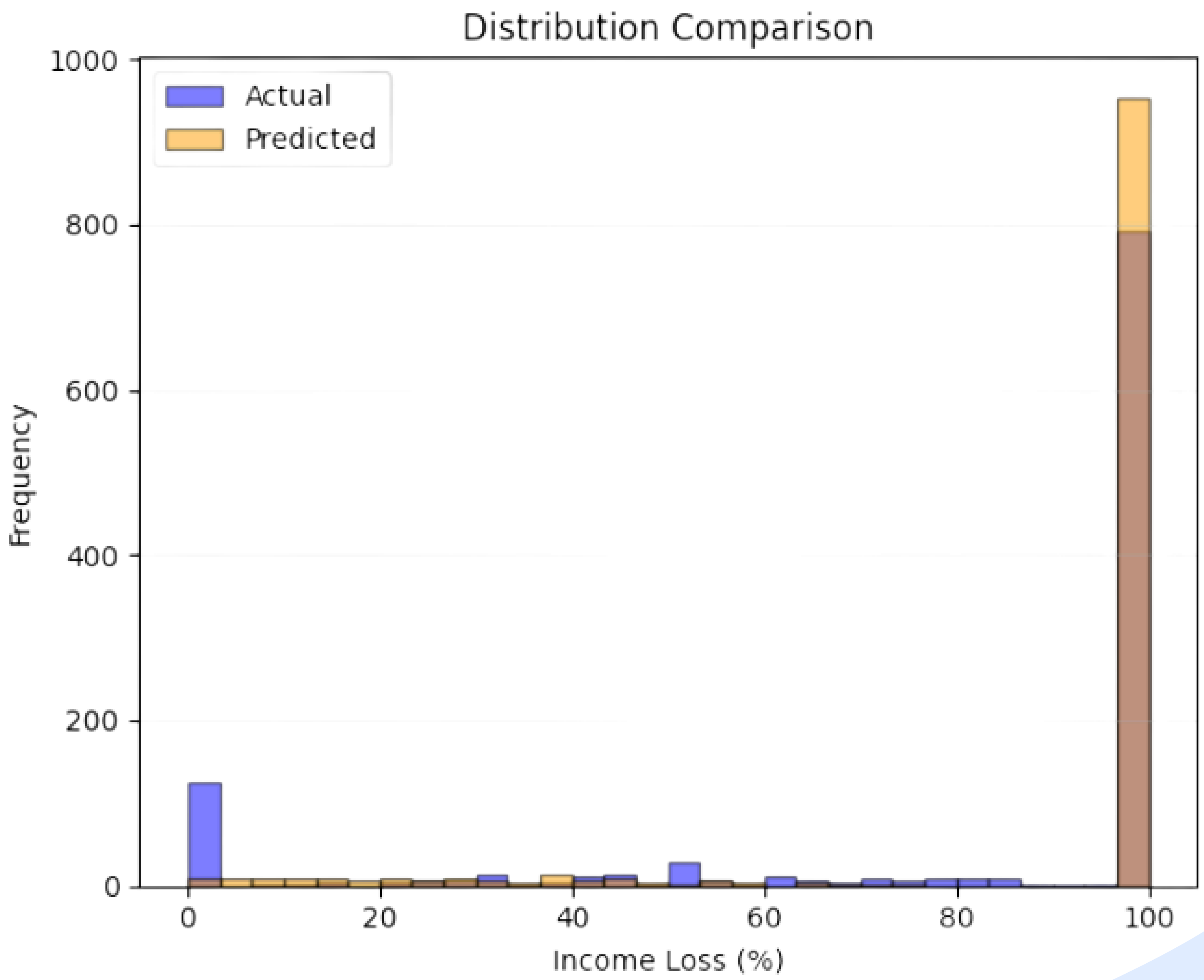
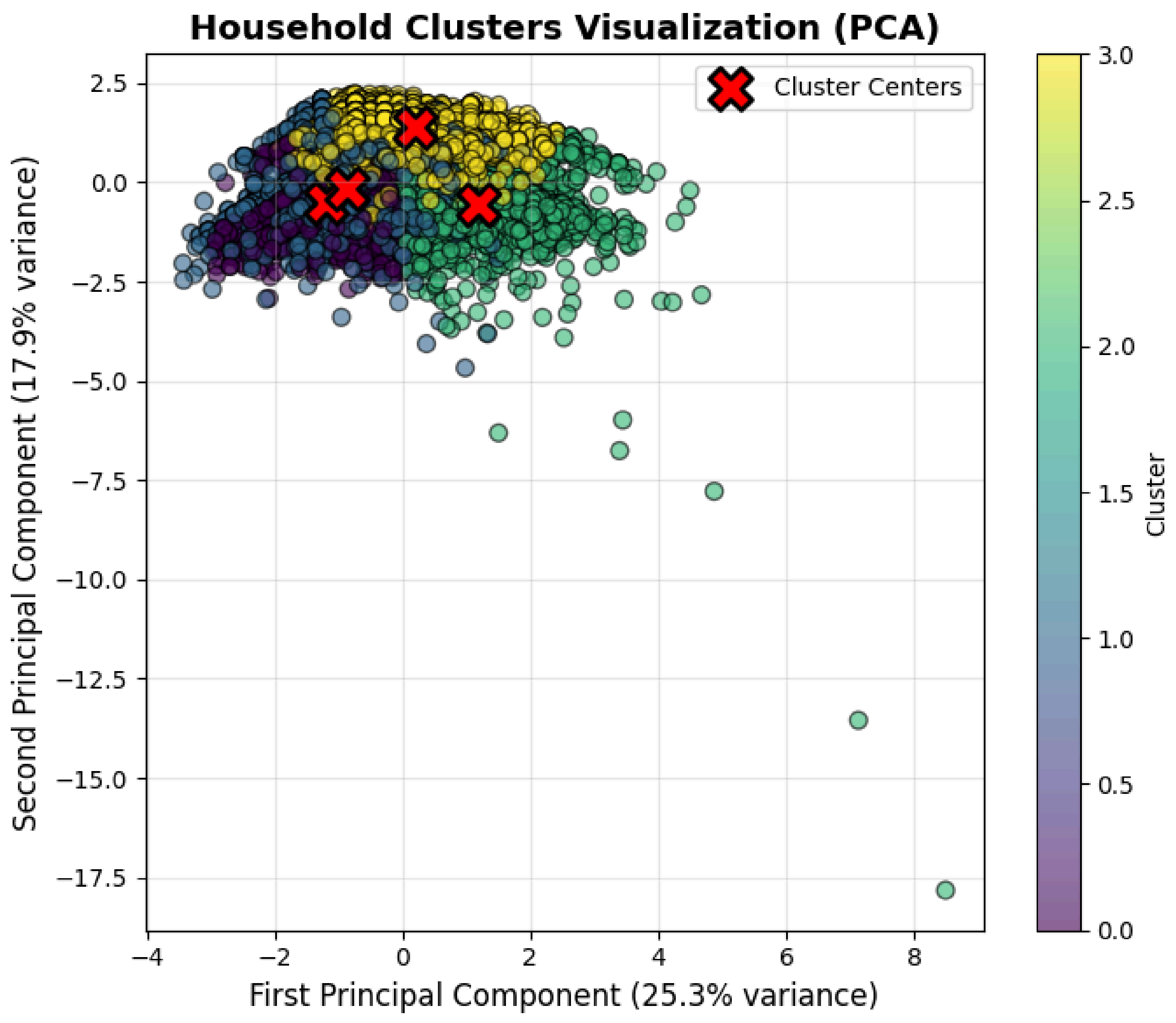
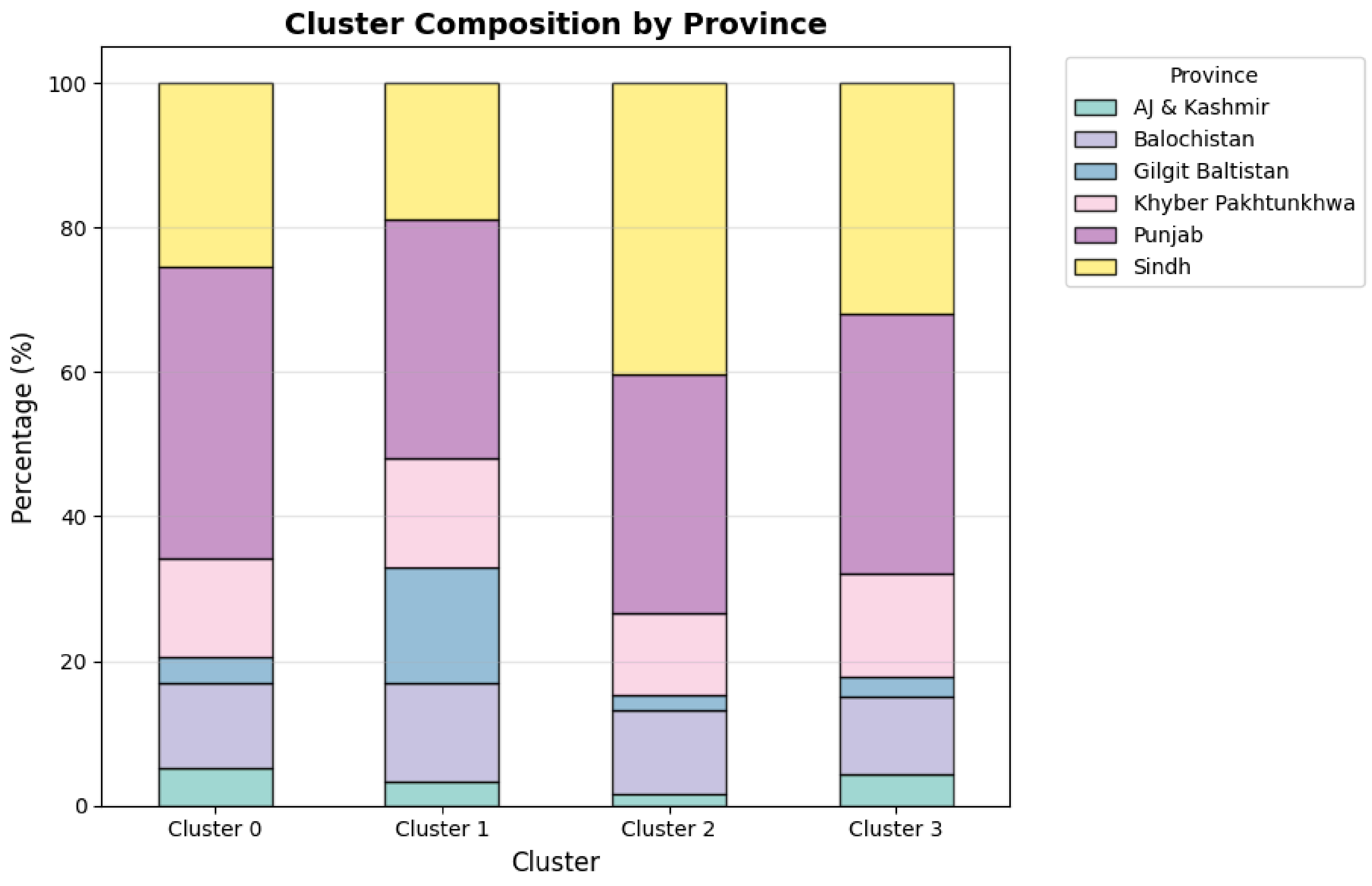


K-MEANS CLUSTERING





- Cluster 0: Low Vulnerability
    - Urban, primarily Punjab
    - Avg income loss: 96.4%
    - Food insecurity rate: 4.0%
    - Education level: **1.89**
- Cluster 1: Low Vulnerability
    - Urban, primarily Punjab
    - Avg income loss: 9.8%
    - Food insecurity rate: 18.0%
    - Education level: 1.13
- Cluster 2: High Vulnerability
    - Primarily Sindh
    - Avg income loss: **95.0%**
    - Food insecurity rate: **65.0%**
    - Education level: 0.50
- Cluster 3: Moderate Vulnerability
    - Rural, primarily Punjab
    - Avg income loss: 93.8%
    - Food insecurity rate: 31.0%
    - Education level: 0.41

SUMMARY & KEY FINDINGS

Consolidated insights from clusters

Cluster 0 — "Economically Shocked but Food Secure" (Low Vulnerability)

Despite severe income loss, households in this cluster remain relatively food secure, likely due to higher education levels, more stable pre-crisis conditions, and stronger social/economic buffers. This suggests that income loss alone does not determine vulnerability, and resilience factors mitigate risk.

Cluster 1 — "Low-Income Urban Households with Moderate Food Stress" (Low Vulnerability)

These households experience stable economic conditions but exhibit moderate food insecurity due to structural disadvantages, particularly lower educational attainment and lower baseline income. Assistance levels are relatively high, indicating targeted government support.

Cluster 2 — "Severely Food-Insecure and Highly Vulnerable Households" (High Vulnerability)

This is the most vulnerable cluster. Households face a compound crisis: major income shocks, high food insecurity, and extensive use of harmful coping mechanisms. Low education and structural inequalities likely intensify vulnerability. The high level of government assistance indicates significant identified need, but outcomes remain poor.

Cluster 3 — "Rural Households Under Moderate Stress" (Moderate Vulnerability)

These households experience substantial economic loss but display intermediate levels of food insecurity, suggesting partial resilience. Rural location and lower education constrain coping capacity, placing them between low-risk and high-risk clusters.

INTERPRETATION & SOCIAL INSIGHTS

- Fairness & Bias Analysis by Province
- Comparative Fairness Analysis: Economic Impact Model
- Key insights
- Additional Analyses for Vulnerable Populations **to understand vulnerability patterns including:**
  - Income loss by type of work
  - Family size vs food insecurity
  - Government/NGO aid coverage for vulnerable households
  - Healthcare access index

Fairness & Bias Analysis by Province

Is there geographical bias in our predictions?

Do models have similar accuracy across all provinces?

Are certain provinces more likely to have false negatives (missing vulnerable households)?

Food insecurity by Province

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FAIRNESS ANALYSIS BY PROVINCE - FOOD INSECURITY PREDICTION
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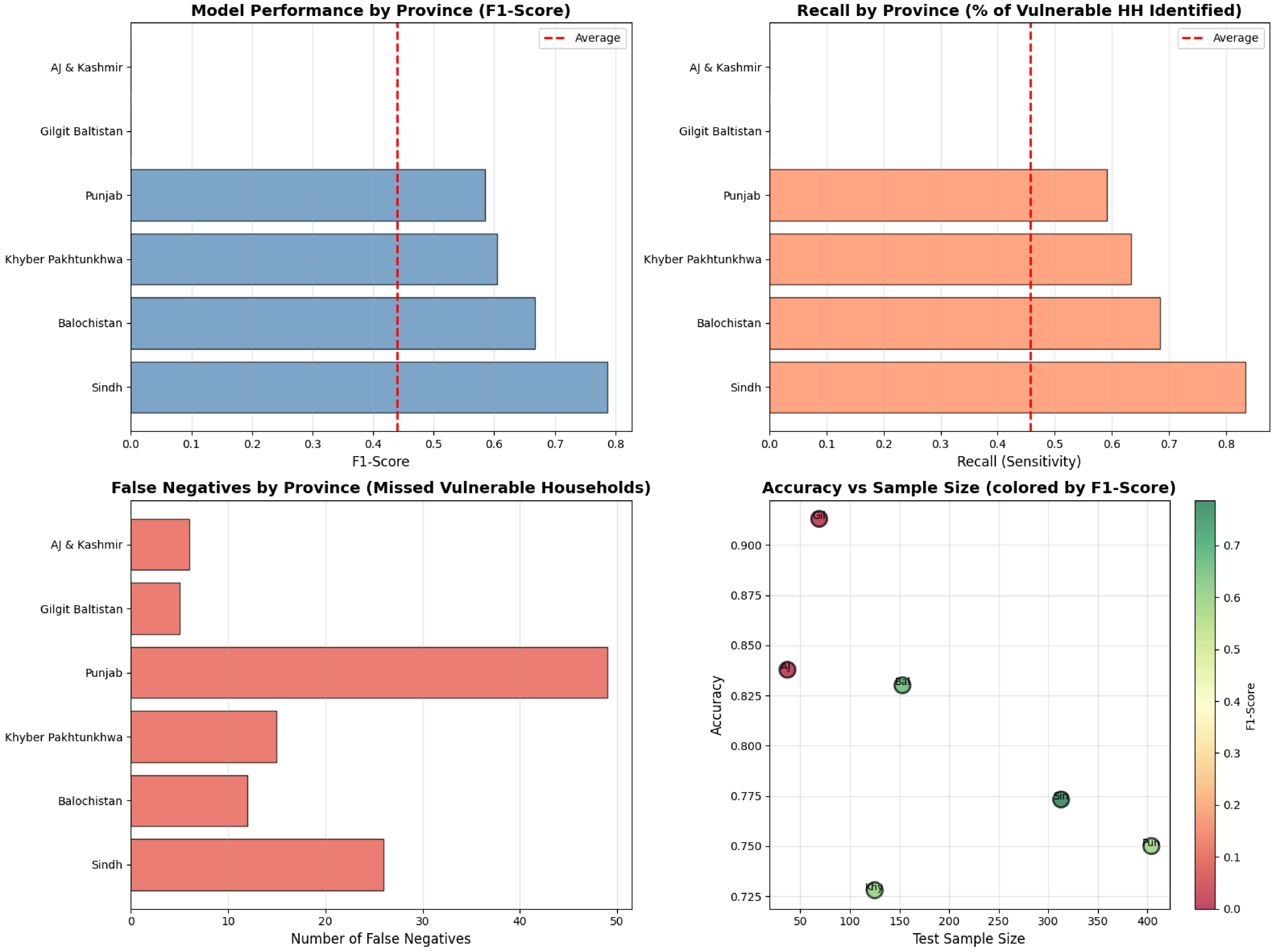
Province	Sample Size	Actual Positive	Accuracy	Precision	Recall	F1-Score	False Negatives	False Positives
Sindh	313	157	0.7732	0.7443	0.8344	0.7868	26	45
Balochistan	153	38	0.8301	0.6500	0.6842	0.6667	12	14
Khyber Pakhtunkhwa	125	41	0.7280	0.5778	0.6341	0.6047	15	19
Punjab	404	120	0.7500	0.5772	0.5917	0.5844	49	52
Gilgit Baltistan	69	5	0.9130	0.0000	0.0000	0.0000	5	1
AJ & Kashmir	37	6	0.8378	0.0000	0.0000	0.0000	6	0

Overall Findings:

Results show major performance disparities, indicating significant geographical bias.

- Sindh is the best-performing province, with the highest F1-score (0.7868) and strong recall (83%).
- AJK and GB perform the worst, with F1 = 0, meaning the model fails to identify any food-insecure households in these regions.
- Missing all positive cases in GB and AJK means the model completely overlooks food-insecure households in these areas, raising severe concerns for equitable policy targeting.
- The disparity indicates that the model is much more accurate in regions with larger, balanced datasets (Sindh, Punjab) and struggles in smaller or underrepresented regions (GB, AJK).


Model Performance by Province (F1-Score)
Average
AJ & Kashmir
Gilgit Baltistan
Punjab
Khyber Pakhtunkhwa
Balochistan
Sindh
F1-Score
Recall by Province (% of Vulnerable HH Identified)
Recall (Sensitivity)
False Negatives by Province (Missed Vulnerable Households)
Number of False Negatives
Accuracy vs Sample Size (colored by F1-Score)
Accuracy
Test Sample Size
F1-Score
Gil
AJ
Bal
Sin
Pun
Khy

Economic Impact Model

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FAIRNESS ANALYSIS BY PROVINCE - ECONOMIC IMPACT PREDICTION
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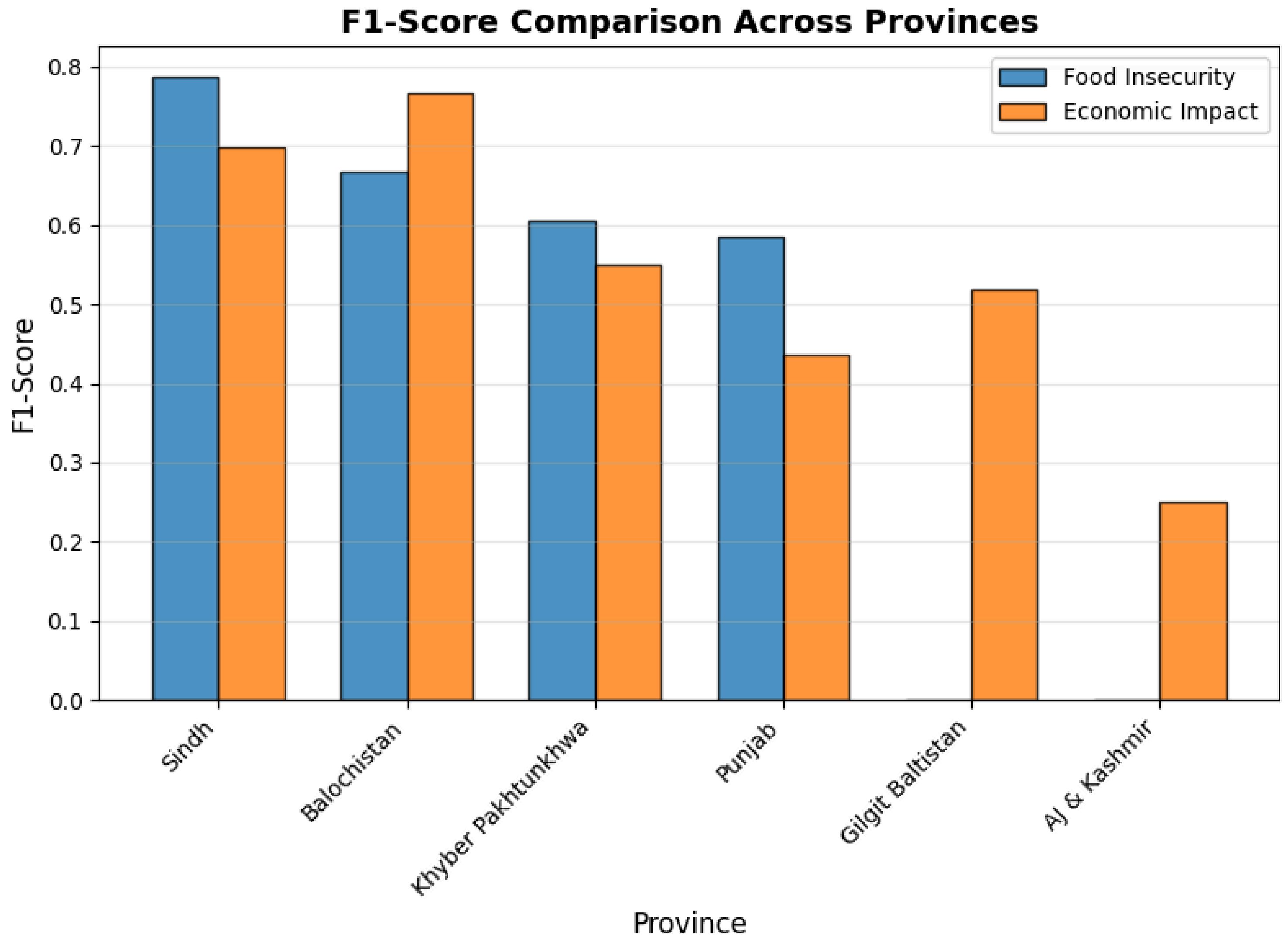
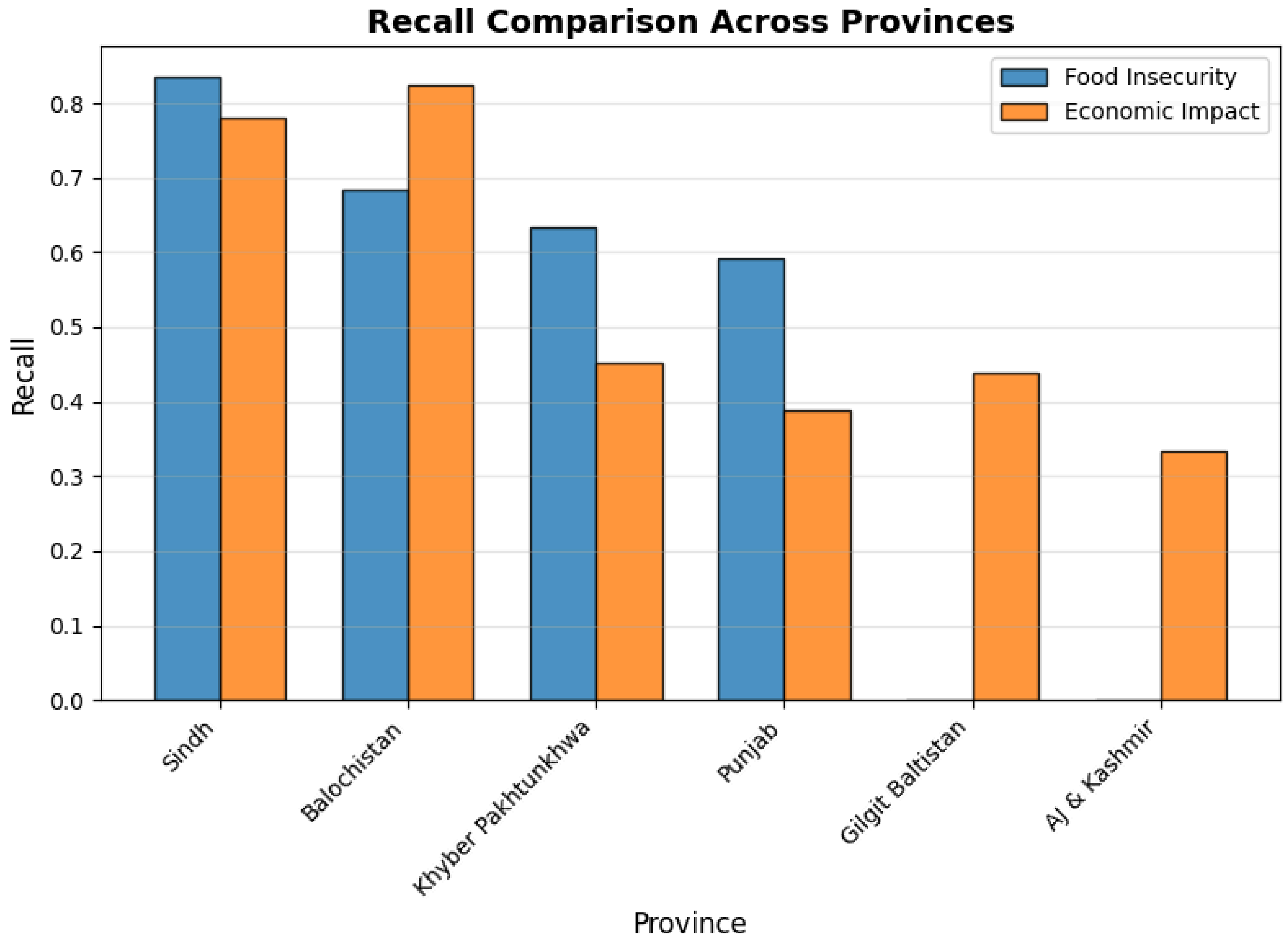
Province	Accuracy	Recall	F1-Score	False Negatives
Punjab	0.8753	0.3878	0.4368	30
Sindh	0.8194	0.7812	0.6977	21
Balochistan	0.8374	0.8250	0.7674	7
Khyber Pakhtunkhwa	0.8369	0.4516	0.5490	17
Gilgit Baltistan	0.7451	0.4375	0.5185	9
AJ & Kashmir	0.8182	0.3333	0.2500	2

- Economic impact = whether the household experienced any financial hardship based on the survey data, such as Income loss, job loss or reduction in working hours, difficulty paying bills or buying essentials, borrowing money or selling assets to cope, Increase in household debt and reduction in savings.
- The model works best for Sindh and Balochistan, but underperforms for Punjab, KP, GB, and especially AJK, indicating substantial geographical bias.

Comparative Analysis

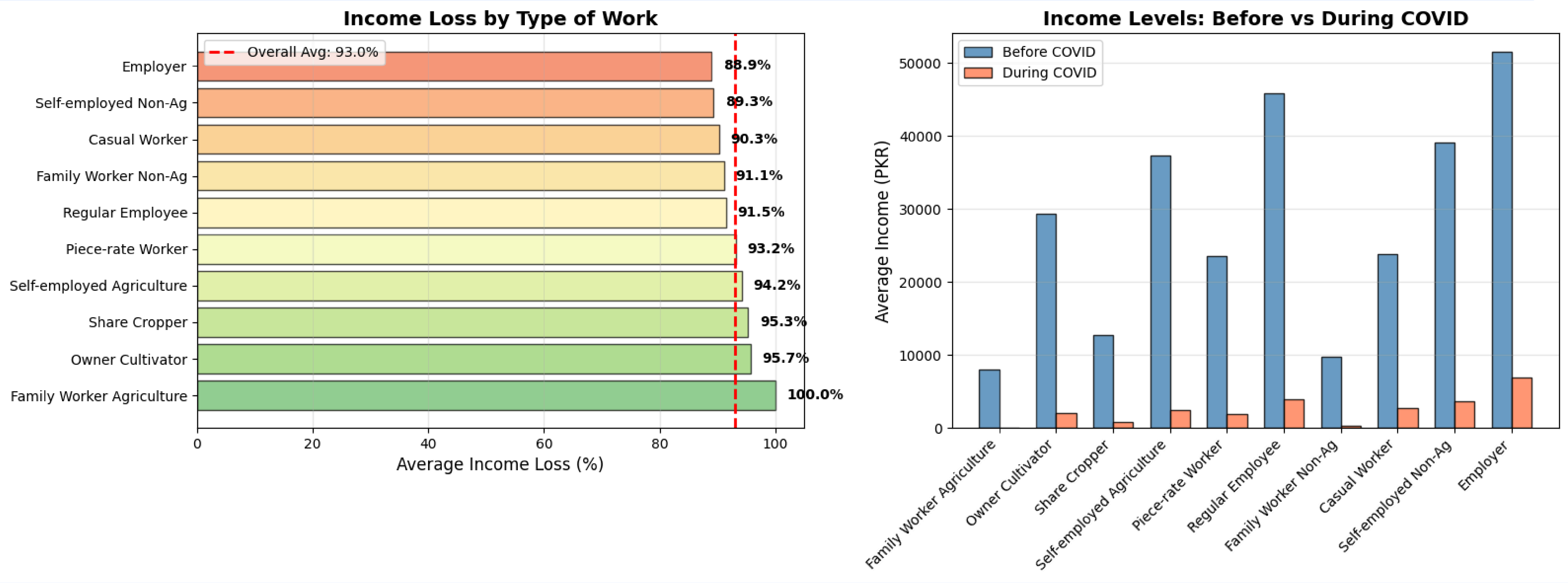




- Sindh and Balochistan show strong model performance for both outcomes, indicating more learnable and consistent patterns in these provinces.
- Punjab, KP, GB, and AJK show weaker performance overall. Their recalls are low showing that the model misses many truly vulnerable households, especially on economic impact prediction.
- Although the food model collapses entirely in data-scarce regions, making its fairness issues more severe in extreme cases.

1. Income loss by type of work

Understanding which types of workers were most affected by COVID-19.

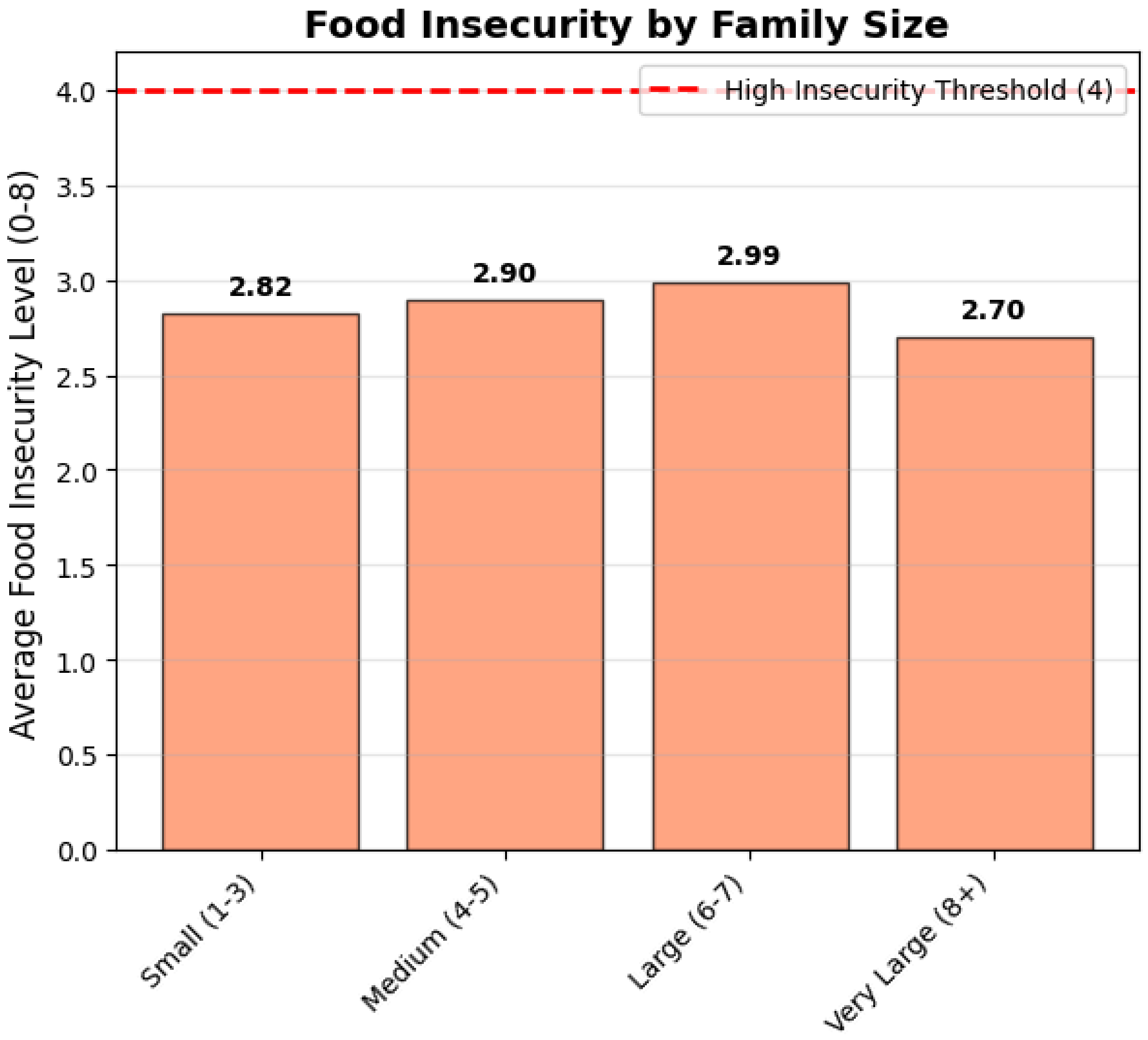


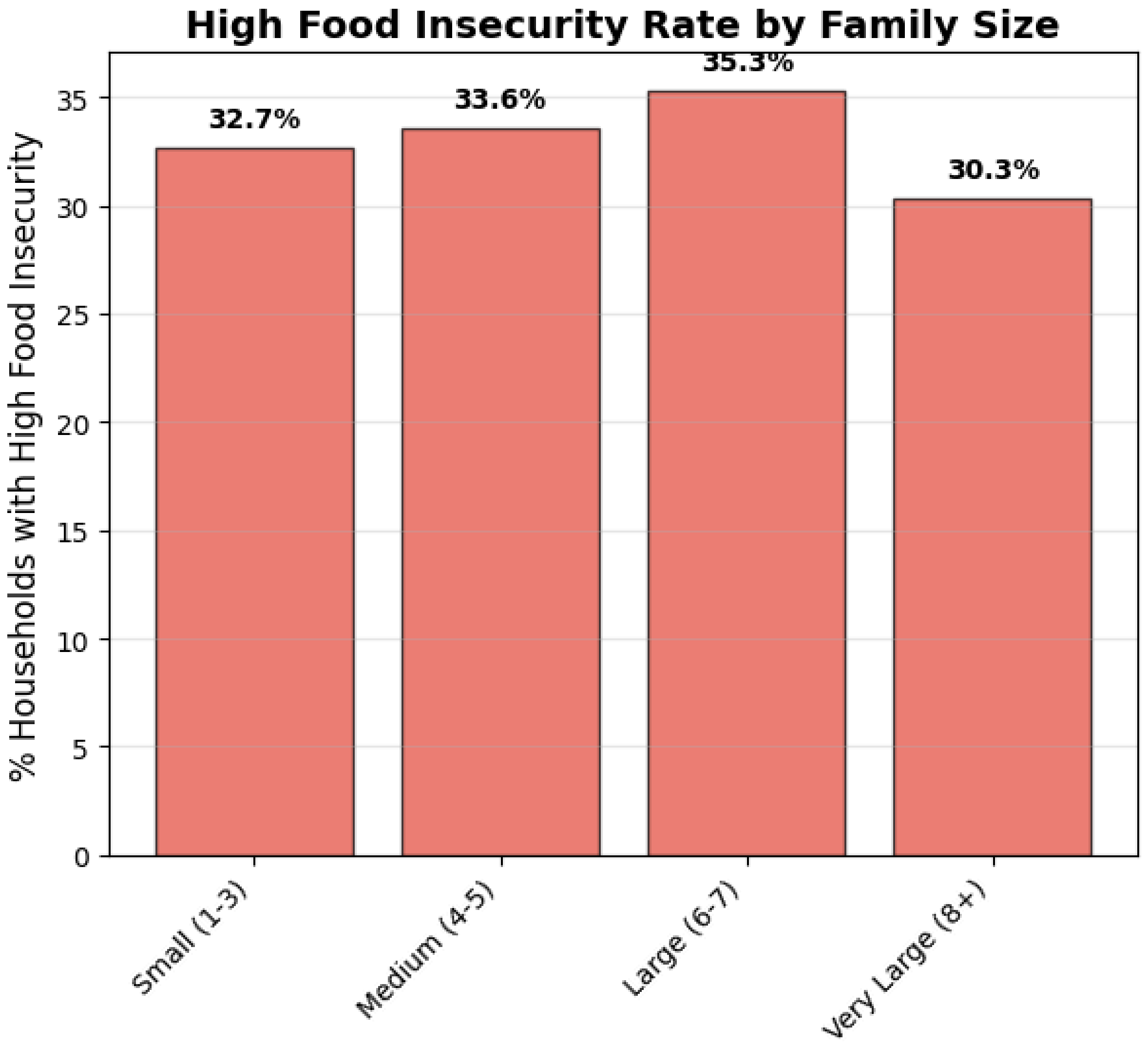
Most Affected: Family Worker Agriculture experienced the highest loss at 100.0%.

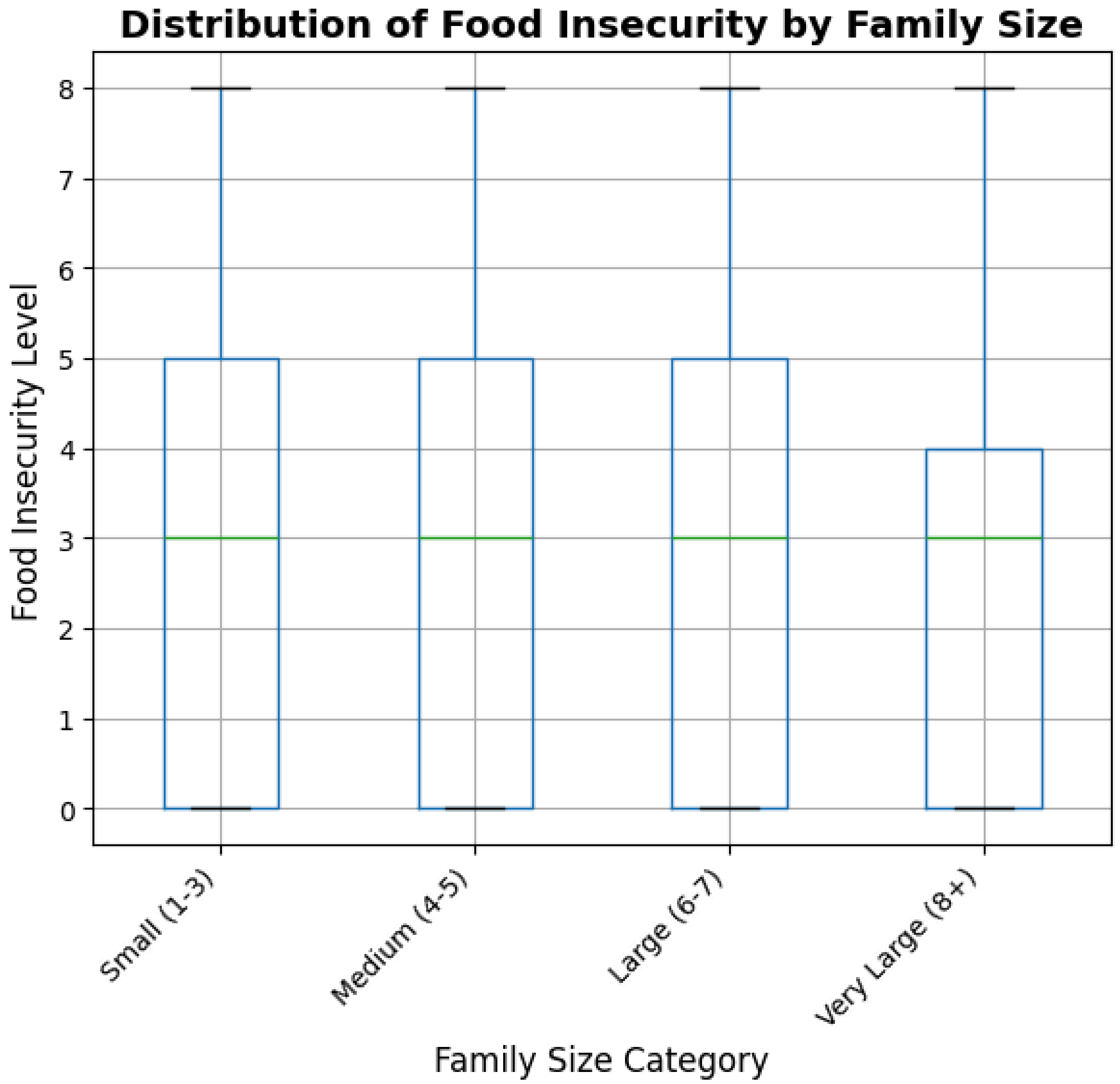
Highest Absolute Loss: Employers and Regular Employees saw the largest income drops in PKR terms, despite slightly lower percentage losses (88.9% and 91.5%).

ANALYSIS 2: FAMILY SIZE VS FOOD INSECURITY

Examining whether larger households experienced higher food insecurity.







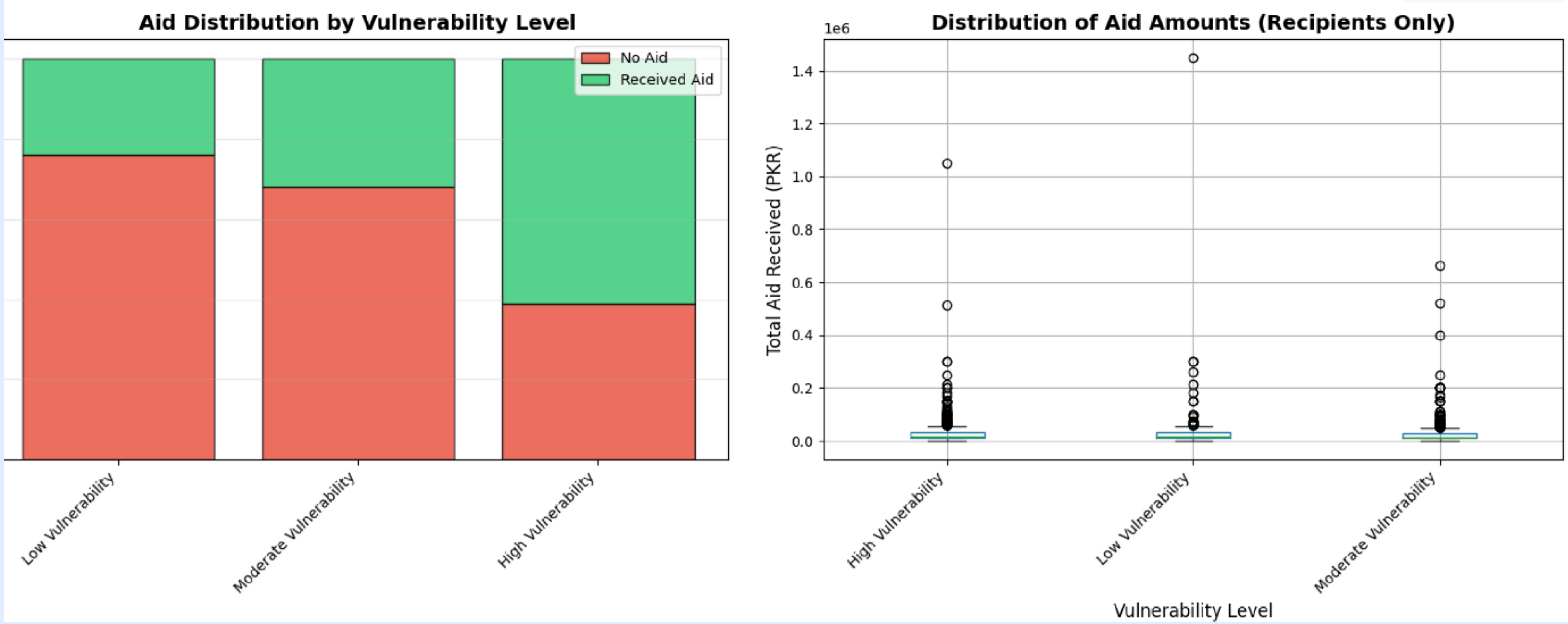
There is a slight trend of increasing average insecurity from Small (2.82) up to Large families (2.99). However, the average level drops for Very Large families (2.70). This specific average measure does not show a simple, linear increase with size.

A significant portion of households in all size categories face high food insecurity

The whiskers and overall range indicate that while the median is low, all family size categories have a substantial proportion of households experiencing the highest levels of food insecurity (up to Level 8).

Analysis 3: Government/NGO Aid for Vulnerable Households

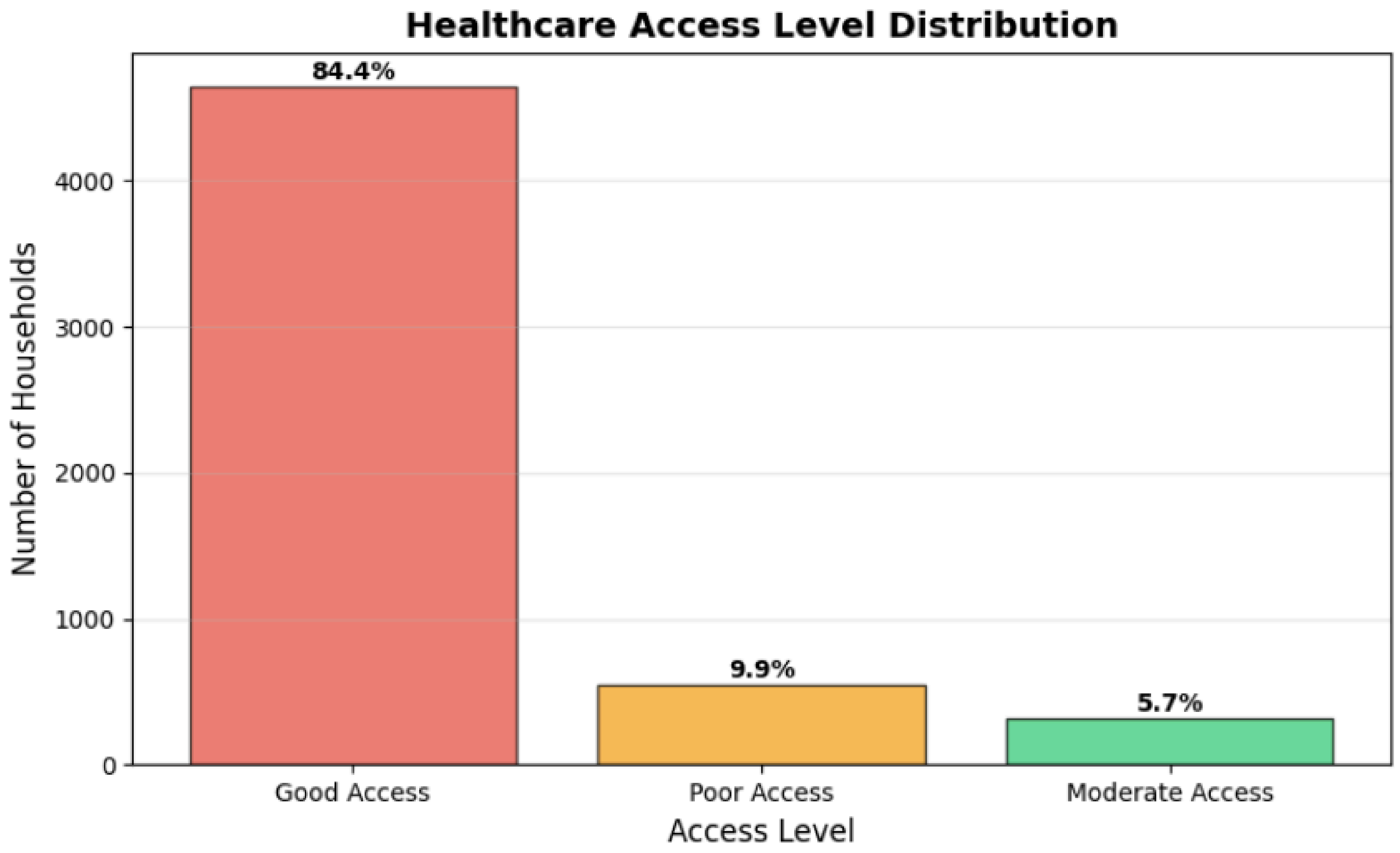
Checking if the most vulnerable households received assistance.



- Higher-vulnerability areas received the most aid, while the lowest-vulnerability areas were the majority among those receiving no aid.
- Similar distribution of aid amounts among recipients across all three vulnerability levels.

ANALYSIS 4: HEALTHCARE ACCESS INDEX

Higher-vulnerability areas received the most aid, while the lowest-vulnerability areas were the majority among those receiving no aid



SOCIETAL REFLECTION

- Agricultural, informal, and even formal workers faced severe income collapse, showing Pakistan's labor market offers employment but little real protection during crises.
- Very large households showed lower food insecurity, highlighting the strength of family and social networks where institutional safety nets are weak.
- Aid is correctly directed toward high-vulnerability areas (e.g., Urban Sindh), but the support is limited, short-term, and insufficient for long-term resilience.
- Low-vulnerability regions like AJK and GB risk policy neglect, even though they are structurally marginal and may be highly exposed in future shocks.

Overall vulnerability is driven by structural inequalities—in employment type, geography, education, and access to protection

POLICY INTEGRATION BASED ON INSIGHTS

Region-Specific, Multi-Dimensional Support for Urban Sindh

Given consistent indicators of highest vulnerability, Urban Sindh requires:

- Integrated food security programs
- Employment stabilization initiatives
- Education and skills development

Transition From Cash-Only Aid to Capability-Building Support

Cash transfers provide temporary relief but do not shift long-term vulnerability.

Policy should incorporate:

- Life-skills and financial literacy training
- Micro-enterprise development
- Job reskilling for urban low-income groups

Design Sector-Specific Safety Nets

For informal and agricultural workers:

- Emergency income protection mechanisms
- Subsidized agricultural insurance
- Guaranteed minimum earning schemes during disasters

For the formal employer–employee sector:

- Establish a mandatory calamity savings fund, separate from insurance
- Monthly contributions from employers and employees
- This ensures that future shocks are met with built-in buffers.

CONT....

Preventive Support for Low-Vulnerability but Marginal Regions

AJK and Gilgit-Baltistan should not be deprioritized simply because the model shows low current vulnerability.
Policy should ensure:

- Preventive food and nutrition support
- Infrastructure upgrades
- Economic diversification

Strengthen Community-Led Support Systems

Since large households show higher resilience, policy can:

- Support community food-sharing networks
- Enhance neighborhood-level welfare committees
- Invest in local support infrastructures

QnA